Streetify Project - Validation Files Analysis by Member

This document tracks and outlines the full project structure specifically identifying where validation logic (both Backend constraints/DTOs and Frontend form/state validations) resides for each module, assigned to the 6 team members.

1. User Account Management & Verification Module

Assigned To: Lahiru (IT25102208)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/AuthController.java (Authentication & registration request validation)
- streetify-backend/src/main/java/com/streetify/dto/LoginDTO.java (@NotBlank, @Email validations)
- streetify-backend/src/main/java/com/streetify/dto/UserRegistrationDTO.java (Passenger form input validation constraints)
- streetify-backend/src/main/java/com/streetify/dto/DriverRegistrationDTO.java (Driver-specific requirements like NIC, License validations)
- streetify-backend/src/main/java/com/streetify/entity/User.java (JPA Entity constraints and database-level validations)

Frontend Validation Files:

- streetify-frontend/src/screens/Login.tsx (Form data parsing, email regex, and password length checks)
- streetify-frontend/src/screens/Profile.tsx (Profile update form validations)
- streetify-frontend/src/components/LahiruUserDashboard.tsx (User component data checks)
- streetify-frontend/src/utils/validators.ts (Reusable client-side validation logic functions)

2. Ride Booking and Dispatch Engine

Assigned To: Chanuka (IT25102207)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/TripController.java (Validates booking payloads before dispatching)
- streetify-backend/src/main/java/com/streetify/dto/TripRequestDTO.java (Coordinate bounds, @NotNull on pickup/dropoff addresses)
- streetify-backend/src/main/java/com/streetify/entity/Vehicle.java (Vehicle properties constraints and driver mapping validation)

Frontend Validation Files:

- streetify-frontend/src/screens/Booking.tsx (Validations ensuring pickup and dropoff fields are set before allowing ride dispatch)
- streetify-frontend/src/components/VehicleTypeSelector.tsx (Ensuring valid ride type selection)
- streetify-frontend/src/components/ChanukaBookingDashboard.tsx
- streetify-frontend/src/OsmMap.tsx (Map-based coordinate validation and bounds checking)
- streetify-frontend/src/hooks/useGeolocation.ts (Permissions validation and GPS availability checks)

3. Trip Progress & Telemetry (Driver Trip Request Mgmt)

Assigned To: Tharindu (IT25102241)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/TripTrackingController.java (Validating live location stream and trip updates)
- streetify-backend/src/main/java/com/streetify/dto/TripStatusUpdateDTO.java (State machine transition validations ensuring valid trip states)
- streetify-backend/src/main/java/com/streetify/entity/Trip.java (Entity-level annotations for required fields during an active trip)

Frontend Validation Files:

- streetify-frontend/src/screens/Driver.tsx (Button disable/enable logic validating if a driver can accept, start, or end a trip)
- streetify-frontend/src/services/tripSyncService.ts (WebSocket incoming message payload validations)

4. Payment and Ledger Management

Assigned To: Daham (IT25102225)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/PaymentController.java (Checking fare consistency and payment status validations)
- streetify-backend/src/main/java/com/streetify/dto/PaymentRequestDTO.java (Ensuring payment amount is valid and method is recognized)

Frontend Validation Files:

- streetify-frontend/src/screens/Payment.tsx (Client-side checks for credit card numbers, wallet balances, and form completeness)
- streetify-frontend/src/components/ReceiptDisplay.tsx (Data validation before rendering the receipt)
- streetify-frontend/src/components/DahamPaymentDashboard.tsx

5. Review, Rating, and Feedback Management (Review & Dispute)

Assigned To: Mithun (IT25102193)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/ReviewController.java (Validating ratings against a completed trip)
- streetify-backend/src/main/java/com/streetify/dto/ReviewSubmitDTO.java (@Min(1) and @Max(5) constraints for star ratings)
- streetify-backend/src/main/java/com/streetify/controller/DisputeController.java (Dispute ticket submission validations)
- streetify-backend/src/main/java/com/streetify/dto/DisputeSubmitDTO.java (Reason/Description @NotBlank and length constraints)
- streetify-backend/src/main/java/com/streetify/dto/DisputeResolveDTO.java (Validating resolution notes)

Frontend Validation Files:

- streetify-frontend/src/screens/Review.tsx (Star rating bounds and comment length validation before submission)
- streetify-frontend/src/screens/Support.tsx (Dispute form validation checking for empty subjects/descriptions)
- streetify-frontend/src/screens/History.tsx (Validating trip history search and filtering logic)

6. Administration and System Control (Admin Governance)

Assigned To: Vidura (IT25102240)

Backend Validation Files:

- streetify-backend/src/main/java/com/streetify/controller/AdminController.java (Role-based access control and privilege validations)
- streetify-backend/src/main/java/com/streetify/dto/SuspendUserDTO.java (Suspension reason and time duration checks)
- streetify-backend/src/main/java/com/streetify/config/GlobalExceptionHandler.java (Centralized hub catching MethodArgumentNotValidException)

Frontend Validation Files:

- streetify-frontend/src/screens/Admin.tsx (Extensive form validations for CRUD operations, driver approval, and trip modifications)
- streetify-frontend/src/components/ModuleExportCard.tsx (Validating requested export formats and date ranges)
- streetify-frontend/src/screens/BranchKiosk.tsx (Validating manual kiosk input data)
- streetify-frontend/src/components/NotificationCenter.tsx (Validating notification types before displaying)